Final Project/Class Reflection

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DSC 250: Data Wrangling for Data Science

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August 10, 2025

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For my final project, I used the same dataset from my midterm but added an important new transformation. The original CSV was stored in a single column with all values separated by semicolons, which made it unusable in its raw state. My first step was to split that single column into multiple fields, assigning proper column names and converting each to the correct data type. This ensured that the dataset could be analyzed effectively and consistently with the transformations I had performed in my midterm project.

With the cleaned and properly structured dataset, I created five new visualizations to uncover different insights. The bar plot showing total stakes by sport revealed where bettors were placing the most money, which could help identify high-interest sports. The pie chart breaking down bet types gave a quick view of betting style preferences, while the scatter plot of odds versus gain highlighted patterns in risk and reward. The box plot of Gross Gaming Revenue (GGR) by sport gave a clearer understanding of revenue consistency and volatility. Finally, the histogram of odds, overlaid with a normal distribution curve, provided a statistical view of how betting odds are distributed, which is useful for modeling and risk assessment.

Through this project, I learned how powerful visualizations can be in revealing trends and patterns that might not be obvious from raw numbers alone. Each chart told a slightly different story, and the markdown explanations helped me articulate those stories in a clear, meaningful way. I also reinforced the importance of thorough data preparation. Without splitting and converting the columns correctly, none of the analysis or visualizations would have been possible.

One of the main struggles I faced was deciding which visualizations would best complement the ones from my midterm without being repetitive. I wanted to provide new perspectives on the dataset, so I focused on different chart types, different focal points, and different analytical angles. Another challenge was ensuring that each visualization wasn't just "nice to look at" but actually conveyed a relevant insight backed by the data.

Overall, this project solidified my understanding of data transformation, visualization, and storytelling through analysis. It also reminded me that while coding skills and statistical methods are essential, the ability to communicate findings clearly is just as important. If I had more time, I would experiment with interactive visualizations to make the insights more engaging for the audience.